SI 612: Milestone 3 [Group 3]

UNIVERSITY OF MICHIGAN

Experience Prototyping Results & Demo Plans

Introduction

At the end of Milestone 2, the project had been narrowed its focus to a specific behavioral challenge: helping students transition out of late-night doomscrolling and into more intentional wind-down routines. We had identified this as a recurring issue in student life, especially within shared living environments where routines are inconsistent and often disrupted. The broader goal was clear. We wanted to explore how subtle, non-intrusive interventions could support behavior change without feeling restrictive or overly controlling.

Several aspects of the project scope had already been defined. We had established three key design dimensions to guide our exploration: the level of control (adaptive versus user-driven), the type of intervention (ambient, tactile, or spatial), and the timing of intervention (arrival home, during active scrolling, or at bedtime). These dimensions helped us systematically generate a wide range of concepts, which were organized into a speed dating matrix.

At the same time, many important details were still unresolved. It was unclear which types of interventions would actually influence behavior in real-world contexts, and when users would be most receptive to support. We also did not yet understand how users would perceive different levels of intervention, particularly in terms of intrusiveness, effort, and social visibility. These open questions shaped the direction of Milestone 3, where the focus shifted from ideation to testing these concepts through user enactments.

Study Design

Describe the plan for your experience prototyping study and the rationale behind your plan. Describe your prototyping setup and scenarios well enough that a reader/listener can easily picture what the experience was like for participants. Give the details of how it was executed (number of participants, study session plans, debriefing strategies, etc.).

To explore how these concepts would play out in realistic settings, we designed a series of experience prototyping sessions using user enactments. The goal was not to test fully functional systems, but to simulate interactions closely enough that

participants could meaningfully respond to them. This allowed us to evaluate how different interventions might feel in context, rather than relying on abstract feedback.

Our study was structured around a set of scenarios derived from the speed dating matrix. Each scenario represented a specific combination of intervention type, timing, and control style. For example, some enactments focused on ambient lighting cues when participants arrived home, while others simulated a phone docking station that gradually increased in prominence during late-night scrolling. Tactile objects were also introduced in certain scenarios to explore whether physical interaction could interrupt scrolling behavior.

The prototyping setup was intentionally simple but carefully staged. We used everyday objects such as lamps, phone placement areas, and small handheld items to represent different intervention types. Changes in lighting, sound, and placement were used to simulate system behavior. Participants were asked to imagine themselves in familiar situations, such as returning home after a long day or scrolling in bed late at night, and to interact with the setup as they normally would.

We conducted these enactments with five participants, all of whom were students living in either dorms or shared apartments. Each session followed a similar structure. Participants were first introduced to the scenario and given minimal instructions to preserve natural behavior. During the enactment, we observed their reactions, noting what they noticed, ignored, or questioned. After each scenario, we conducted a short debrief to understand their thought process, including what felt intuitive, what felt awkward, and whether the intervention would realistically influence their behavior.

This approach allowed us to capture not only immediate reactions, but also deeper reflections on usability, intrusiveness, and real-world applicability. By grounding our study in enacted experiences, we were able to generate insights that directly informed the evolution of our design direction.

Study Results

Researcher	UE Code	Note
Andy	UE1	UE1-P1 Would notice more if the light color is not normal.
Andy	UE1	UE1-P1 Might notice the light but not immediately connect it to an action.
Andy	UE1	UE1-P1 Would the participant stop scrolling when seeing the light? Maybe, not sure.
Andy	UE1	UE1-P1 Would respond more if the light feels tied to a clear action.
Andy	UE2	UE2-P1 Would not actively look for tactile projects if they want to stop scrolling.
Andy	UE2	UE2-P2 Does not think a tactile object would stop them from scrolling if they have to activey look for them.
Andy	UE2	UE2-P2 Would be more likely to use it if it interrupts naturally without thinking.
Andy	UE2	UE2-P2 Might notice the object but not feel motivated to touch it.
Andy	UE2	UE2-P2 Would interact only if it feels effortless and immediate.
Andy	UE3	UE3-P1 Still feels a bit of effort even for small actions when very tired.
Andy	UE3	UE3-P2 Might ignore it if they feel mentally drained already.
Andy	UE3	UE3-P3 Likes the idea of doing just one simple thing instead of a full routine.
Andy	UE3	UE3-P3 Would follow through more if the action feels quick and easy.
Skylar	UE4	UE4-P4 questioned what happens if they place the phone down but then want to pick it up again shortly after; this suggests potential friction in real usage and unclear interaction flow.
Skylar	UE4	UE4-P4 was concerned about the tone of reminders; if the reminders feel too annoying or intrusive, they may choose to turn off the device entirely.
Skylar	UE4	UE4- P4 wondered whether there is a companion mobile app to track usage time or behavior, or if the product functions only as a standalone device.
Skylar	UE4	UE4-P4 asked about the presence of a reward or incentive system (e.g., similar to Forest) to motivate consistent behavior; they also mentioned that gentle reminders alone may not be effective for them.
Skylar	UE4	UE4-P4 questioned the level of user control over the system (e.g., ability to mute or turn off reminders); they pointed out that too much flexibility might reduce the product's effectiveness and perceived value.
Andy	UE4	UE4-P3 Could be more pushy so the participant will notice more.
Andy	UE4	UE4-P3 Would be better if the phone station gradually makes more noticable ambient/sound over time.
Andy	UE4	UE4-P3 Would be more effective when used late at night to help them get out of scrolling in bed.
Andy	UE5	UE5-P5 Would notice even less in a shared space with other people around.
Andy	UE5	UE5-P5 Would avoid interacting if it draws attention from others.
Andy	UE5	UE5-P5 Would prefer something more private and less noticeable in shared settings.

fig 01: User Enactment Notes

The user enactments helped us move from a broad set of conceptual possibilities to a more grounded understanding of what kind of intervention could realistically support students during late-night doomscrolling. Across the sessions, participants did not simply react to whether an intervention was present. What mattered much more was whether it felt clear, low-effort, and acceptable within the social and emotional context of nighttime routines. Rather than just telling us which ideas were interesting, the study helped define what kind of support users would actually tolerate and act on.

One of the clearest findings was that **noticing a cue does not necessarily lead to action**. In the ambient light scenarios, participants often said they would notice a change in lighting, especially if the color felt unusual or different from their normal setup. However, several also suggested that simply noticing a light would not automatically make them stop scrolling. The light could attract attention, but it did not always communicate the next step.

This showed us that:

- awareness alone is too weak as a standalone intervention
- passive ambient cues can be noticed without changing behavior
- the intervention needs to imply a specific action, not just signal a mood shift

We also learned that **effort is a major barrier**, especially when users are already tired or mentally drained. In the tactile object enactments, participants repeatedly suggested that they would not go out of their way to interact with a separate object if they were deep in a scrolling loop. Even if the object was visible, it often felt too easy to ignore because it required an extra deliberate step.

That led to an important distinction:

- a **standalone tactile object** felt too effortful
- a **physical docking action** tied to an existing bedtime routine felt much more plausible

This was important for our design direction. The enactments did not reject physical interaction altogether. Instead, they suggested that physical interaction works better when it is integrated into a natural routine, such as placing a phone down to charge for the night.

Another major insight was that **gradual escalation felt more promising than static reminders**. In the phone station scenarios, participants responded more positively to the idea of a cue that became more noticeable over time rather than one that appeared all at once. Several comments suggested that a reminder might work better if it began subtly and then slowly increased in prominence, especially later at night when the user was already aware that they should be winding down.

This suggested that the system should be:

- gentle at first
- more noticeable only if needed
- responsive to ongoing behavior rather than static

The enactments also supported the idea that **sound could help, but only under certain conditions**. Some participants felt that a sound cue might be effective in pushing them to disengage from scrolling, especially if it emerged gradually rather than abruptly. At the same time, there was clear concern about reminders becoming annoying or overly intrusive. Participants questioned what kind of sound the system would make, whether it could be muted or adjusted, and whether an irritating tone would simply lead them to disable the system.

This revealed an important design opportunity:

- sound should be **secondary**, not the first cue
- it should be **gentle and gradually introduced**
- users should have **limited customization** over tone or reminder style
- the system should support, not punish

The study also made it clear that **social context matters**. In the shared-space scenarios, participants said they would be less likely to engage with an intervention if it drew attention from roommates or others nearby. This meant that louder or more visible interventions had a narrower window of acceptability. Privacy and subtlety were repeatedly valued.

In practice, participants seemed more comfortable with systems that were:

- discreet
- low-visibility
- unlikely to create embarrassment or social awkwardness

Finally, participants raised questions about **interaction flow and ongoing use**, which helped us think beyond a single reminder moment. For example, some wondered what would happen if they placed the phone down and then wanted to pick it up again shortly after. Others asked whether there might be a companion app, some kind of feedback loop, or a motivating incentive structure. These questions suggested that users want the system to feel coherent and flexible rather than rigid or one-dimensional.

Taken together, these findings helped us narrow the design space significantly. The most promising intervention was not a purely ambient object, a separate tactile

product, or a harsh reminder system. Instead, the enactments pointed toward a concept that combines:

- a **clear physical bedtime action**
- **ambient light as an initial cue**
- **gradual escalation over time**
- **optional customizable sound** for users who want a stronger nudge
- **a form factor that remains subtle** and compatible in shared living spaces

This directly led us toward a bedside wind-down dock: a charging stand that acts as a designated nighttime resting place for the phone and gently guides the user toward putting it down.

Ideation and Selection

1. Ideation Approach

We approached this round of ideation using a structured matrix based on three design dimensions: control style, type of intervention, and intervention moment. This helped us systematically explore how different forms of support might work across moments such as arriving home, active scrolling, and the transition to bedtime. We generated ideas across ambient, tactile, and spatial categories, each offering a different balance of visibility, effort, and user control.

As the enactments progressed, the matrix became more than just a concept generation tool. It also helped us compare ideas based on how they actually felt in context. Through that process, we began to see patterns in what participants responded to, what they ignored, and what they found too effortful or intrusive.

2. Criteria and Tradeoffs

Based on our findings, we refined our evaluation criteria. The strongest concepts were the ones that balanced practical usability with emotional and social acceptability.

We prioritized:

- **usability**: the action should require little effort, especially in low-energy states

- **comfort and acceptability**: the system should not feel annoying, controlling, or embarrassing in shared spaces
- **feasibility**: the concept should be realistic to prototype with accessible technology
- **effectiveness**: the intervention should lead toward a clear behavioral outcome, not just passive awareness

We also identified several important tradeoffs:

- stronger reminders may be more effective, but they also risk becoming irritating
- more user control improves comfort, but too much flexibility may reduce the system's impact
- purely ambient interventions are subtle, but may be too easy to ignore
- purely physical or tactile interventions can be actionable, but may fail if they require too much effort or feel disconnected from the user's routine

3. Final Selection

Using these criteria, we narrowed our concepts toward solutions that combined a clear physical action with a cue that could become more noticeable over time. Passive ambient signals alone often seemed too easy to ignore, while more forceful reminders risked feeling annoying or overly controlling. This pushed us toward a middle ground: a system that remains gentle at first, but can gradually increase in prominence if needed.

Our final proposal is a **bedside wind-down dock**, a phone charging stand that acts as a designated nighttime resting place for the phone. At a chosen point in the evening, the dock begins with a subtle light cue to signal the start of the wind-down period. If the user continues scrolling, the cue gradually escalates through brighter light and, if enabled by the user, a gentle customizable sound reminder. This creates a clear bedtime action, placing the phone onto the dock, while still allowing the experience to feel supportive rather than rigid.

We selected this concept because it reflects several of the key lessons from the enactments:

- users responded better to interventions that implied a **concrete physical action**

- gradual escalation felt more promising than static reminders
- the tone of reminders mattered, which pointed toward the value of **limited customization**
- low-effort, socially subtle interactions felt more acceptable than loud or highly visible interventions

Overall, this concept gave us the strongest balance of usability, effectiveness, comfort, and feasibility within the design space we explored.

System Proposal

Our proposed system is a **bedside wind-down dock** that combines a phone charging stand with ambient light cues and optional sound-based escalation. The system is designed to support the transition from late-night scrolling to sleep by making one specific bedtime behavior easier and more legible: placing the phone onto the dock and leaving it there.

As shown in the storyboard, the dock first signals the start of the wind-down period through ambient light, then gradually becomes more noticeable until the user places the phone on the dock. The dock is meant to function as more than just a charger. It becomes a small part of the bedtime routine, a physical anchor that helps turn the intention to stop scrolling into an actual action. Rather than blocking access to the phone or enforcing a rigid behavior, the system provides a gentle structure for disengagement.

At a high level, the system works like this:

- the user sets a preferred wind-down time or activates bedtime mode
- when that time arrives, the dock begins emitting a soft ambient light
- if the user continues using the phone, the cue slowly becomes more noticeable
- if the user has enabled sound, the system can add a gentle customizable audio reminder
- once the phone is placed onto the dock, the system shifts into a calm resting state and charging begins

fig 02: Storyboard of the Bedside Wind-Down Dock

This interaction model reflects the main themes that emerged from our enactments. The ambient light provides an initial low-pressure signal. The gradual escalation helps the cue remain noticeable without feeling immediately intrusive. The dock itself creates a clear destination for the phone, reducing ambiguity around what the user is supposed to do next.

Core components

A plausible near-term version of the system could include:

- a **charging dock or stand** that acts as the primary physical interface

- a **phone presence sensor** or weight/contact sensor to detect when the phone has been placed down
- **integrated LEDs** for subtle light-based cues and escalation
- a **small speaker or audio module** for optional sound reminders
- a **basic timing and control system** to manage activation and escalation
- an optional **mobile settings interface** for selecting bedtime time, sound type, or intensity preferences

This concept is directly shaped by the design constraints surfaced in the study:

- it is **low-effort**, which matters because users are often tired and cognitively drained at night
- it is **physically grounded**, making the action more concrete than a standard phone notification
- it can remain **discreet**, which is important in dorms and shared apartments
- it supports **augmentation rather than automation**, helping the user follow through on an intention instead of taking full control away from them

We also see the concept as realistic in the near term. A working prototype could be built today using accessible components such as a microcontroller, LEDs, a simple sensor, and a charging base. A more refined version in the future could include smarter detection, more nuanced escalation behavior, and lightweight habit tracking, but the core interaction does not depend on speculative technology. Its main value lies in the experience design.

Demo Proposal

To demonstrate the system, we will create a communicative experience prototype centered on the core bedtime interaction our concept is designed to support: noticing the wind-down cue, experiencing the gradual escalation, and placing the phone onto the dock. The goal of the demo is not to present a fully finished commercial product. Instead, it is to make the intended user experience clear and believable enough that viewers can understand how the system would function in practice and why it emerged from our enactment findings.

Our demo will simulate a realistic late-night scenario in which a user is scrolling on their phone near their bed or desk. The dock will be positioned beside them as part of the environment rather than introduced as a separate object. At the start of the wind-down period, the dock will emit a soft ambient light to signal that it is time to transition toward sleep. If the user continues using the phone, the light will gradually become more prominent. In a later stage, if enabled, the system will also introduce a gentle sound reminder. Once the user places the phone onto the dock, the system will shift into a calm resting state, visually indicating that the intended bedtime action has been completed.

The demo will focus on simulating several aspects of the experience:

- the beginning of the wind-down period
- the gradual escalation of light over time
- the optional addition of a customizable sound cue
- the physical action of placing the phone on the dock
- the calm resting state once the phone has been docked

At least one part of the prototype will be implemented using computational media. We plan to use a microcontroller (Photon) based setup to control the dock's lighting behavior and respond to phone placement through a simple sensor or triggered input. If feasible, we will also incorporate a basic audio module so the prototype can demonstrate how an optional sound reminder would work as a secondary cue. This level of interactivity is important because it allows the demo to communicate not just the appearance of the system, but also its timing and responsiveness.

There are clear reasons why we are not prototyping the system at a higher fidelity at this stage. A more polished and fully functional version would require more advanced sensing, stronger integration with actual phone usage patterns, and a longer development timeline than is realistic for this milestone. It would also require more refined hardware construction and more extensive testing across repeated nighttime routines. Because of those constraints, some aspects of the final concept will remain simulated rather than fully built. These include longer-term adaptation, more sophisticated habit tracking, and any companion settings interface.

Even with those limitations, this demo should preserve the most important experiential qualities of the design. It will show the dock as a physical destination for the phone, demonstrate the difference between a subtle cue and a gradually escalating one, and communicate the intended balance between support and user control. What it will not fully capture is long-term habit formation, since participants will only experience the system in a short demonstration rather than over multiple nights. Still, the prototype will be effective for showing why this concept is plausible, how it fits the findings from our enactments, and what aspects of the experience would matter most in a future higher-fidelity version.

Conclusion

By the end of this milestone, the project has shifted from exploring many possible interventions to a much clearer concept direction. The user enactments helped us narrow the design space and identify a solution that feels both realistic and acceptable in the context of student bedtime routines. Our project is now centered on a **bedside wind-down dock**: a phone charging stand that uses subtle light cues, gradual escalation, and optional customizable sound to help users place their phone down and transition toward sleep.

Several parts of the project are now much more defined. We found that the intervention should be:

- **low-effort**, since users are often tired at night
- **action-oriented**, not just awareness-based
- **gradual**, rather than abrupt or aggressive
- **discreet**, especially in shared living spaces

The enactments also clarified the **cultural boundaries of acceptance**. Participants were most comfortable with systems that felt private, subtle, and supportive. They were less comfortable with reminders that felt embarrassing, loud, overly visible, or too controlling. This suggests that the role of technology in this subcontext should be to **augment**, not automate. The system should help users follow through on an existing intention to stop scrolling, rather than forcing the behavior or taking control away from them.

Our findings also point to the most appropriate interface mix. A purely screen-based solution would be too easy to ignore, and a purely sound-based system would feel too intrusive. The strongest combination appears to be:

- **spatial interaction**, through the dock itself
- **ambient interaction**, through light-based cues
- **optional audio**, only as a secondary escalation
- **minimal screen interaction**, only for setup or customization

Some uncertainties still remain. We do not yet know how well the system would work over long-term use, how much customization users would want, or what the ideal balance between light and sound should be. Still, those questions now feel like refinements rather than major conceptual unknowns.

Overall, the enactments gave us a much clearer direction. They showed that ambient light alone was too weak, separate tactile interaction required too much effort, and stronger reminders needed to stay gentle and user-sensitive. That is what led us to the wind-down dock: a concept that balances guidance, subtlety, and user control in a way that fits the realities of bedtime doomscrolling.

Contribution Report

Skylar Song: I created the document and conducted 2 User Enactments and one User Enactment in class. For the documentation I'm responsible for the ideation and demo proposal part.

Mustafa Ali Mirza: I wrote the Introduction, Ideation + Selection, Study Design and Conclusion sections and made the storyboard diagram and cover page using figma.

Andy Xu: I conducted 3 User Enactments, organized notes spreadsheet and wrote the Study Result and System Proposal sections.